

4. Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets.

Aim

To build an Artificial Neural Network using the **Backpropagation algorithm** and test it using an appropriate dataset.

Software Required

- Python 3.x
- NumPy
- Scikit-learn

Procedure

1. Load the Iris dataset.
2. Divide the data into training and testing sets.
3. Normalize the input features.
4. Create a neural network using the MLP classifier.
5. Train the network using backpropagation.
6. Predict the test data.
7. Calculate accuracy.

Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score

iris = load_iris()
X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

scaler = StandardScaler()
```

```
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
model = MLPClassifier(
    hidden_layer_sizes=(10,),
    max_iter=1000,
    random_state=42
)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Predicted Values:")
print(y_pred)
print("\nAccuracy:",
      accuracy_score(y_test, y_pred))
```

Output:

```
... Predicted Values:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]

Accuracy: 1.0
```

Result

Thus, an Artificial Neural Network using **backpropagation** was successfully implemented and tested.